

CS105.3 Faculty of Computing, Online Examinations 2021

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COMMENTS	

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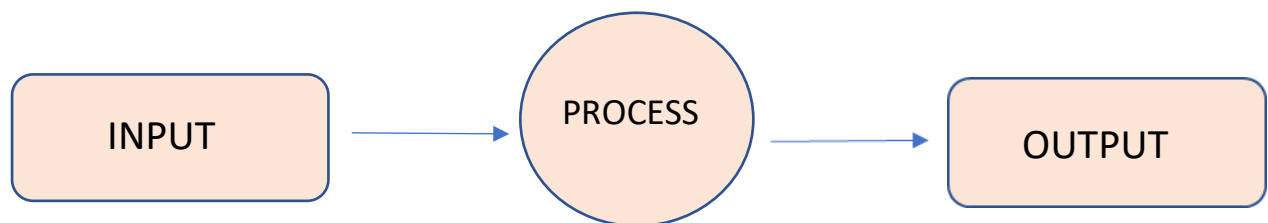
Answers

Question 01

(01)

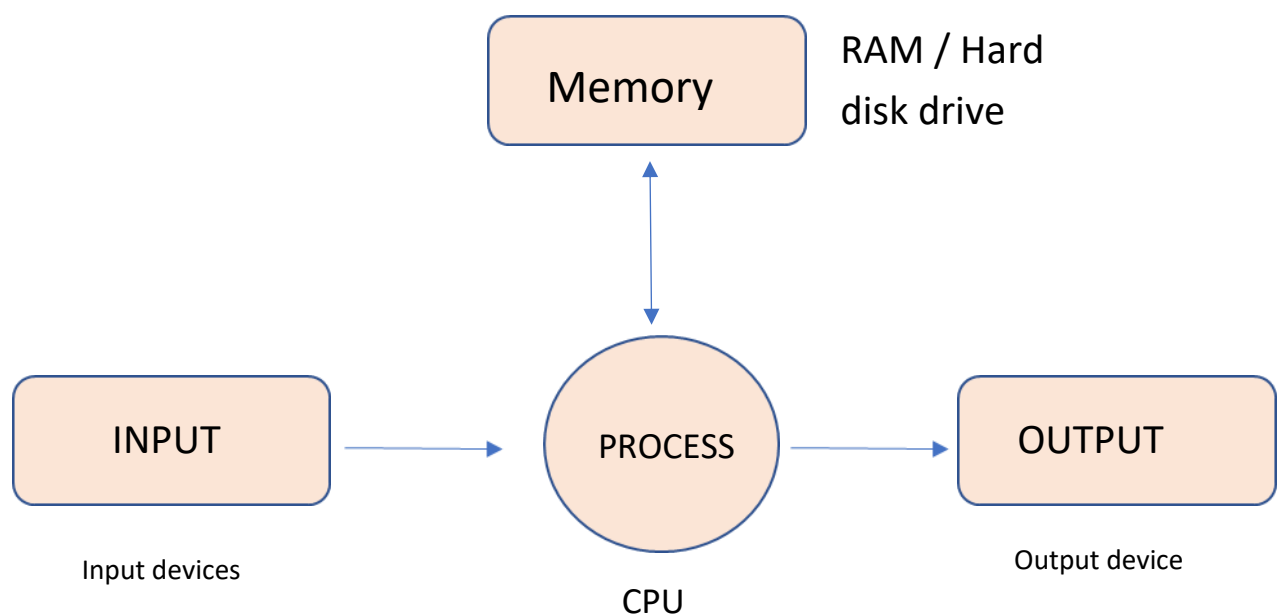
1) Architecture of a typical machine

When we are consider about Typical machine ,those machines are capable of providing one certain output.



2) Architecture of a Computer

Computer is a different type of machine which has, Speed, Versatility, Intelligent, Memory. When come to the processing part computer has two way communication between its memory.



(02)

A central processing unit is an electronic circuitry within a computer that executes instructions that make up a computer program. The CPU performs basic arithmetic, logic, controlling, and input/output operations specified by the instructions.

Types of Processor

- I. Single core processor
- II. Dual core processor
- III. Quad core processor

Single core processor

It is a microprocessor with a single core on a chip, operating a single thread at any one time.

Dual core processor

It contains two processors or two execution cores in the same combined circuit.

Quad Core processor

It is a chip with four individual units called cores.

(03)

System software

We also called it as Operating System which helps to boot up the basic program automatically as soon as the computer started. These Softwares are capable of combining all hardware component together and work as a one unit.

Application Software

These softwares are known as third party applications which running on top of the Operating system. These application softwares are dedicated for a certain operating system. The reason for that is, the every application softwares are running on top the operating system.

(04)

Hard disk Drive (HDD)

It is a non-volatile data storage. It is usually installed on the inside in a computer, attached directly to the disk controller of the computer's motherboard. It connected to the motherboard using an ATA, SCSI or SATA cables. They are powered by a connection to the computer's Power Supply Unit

Solid State Drive (SSD)

SSD is a new invention of storage device used in computers. SSDs replace old mechanical hard disks by using flash based memory, which is significantly faster. SSDs speed up computers significantly due to their low read access times and fast throughputs.

HDD	SSD
Storing data inside a platter/disk.	There are small chips to storing data(integrated chips).
Need to be connected to the mother board.	Need to be connected to the mother board.
Little bit slow.	Speed than HDD.
Lower lifespan.	Longer lifespan.
More heavier than SSD.	Light weight device.

Question 02

(01)

Octal Number System

Mostly used in minicomputers. It says that it is a number system of base 8 which means that we require 8 different symbols in order to represent any number in Octal system. This number system it's capable of converting an Octal number directly to a binary number.

Hexadecimal Number System

In this number system there should be sixteen different symbols to represent numbers hence this number system is quite complicated when considering with other number systems. The base value should be written as 16 .

(02)

To octal

Convert to Octal

(a) 111010101_2

111	010	101
↓	↓	↓
7	2	5

$= \underline{\underline{725_8}}$

(b) 341_{10}

$= \underline{\underline{525_8}}$

8		341	
8		42	- 5
		5	- 2

(c) 10100010100101_2

010	100	010	100	101
↓	↓	↓	↓	↓
2	4	2	4	5

$= \underline{\underline{24245_8}}$

To Hexadecimal

Convert to hexadecimal.

(a) 111010101_2

0001	1101	0101
↓	↓	↓
1	D	5

$= 1D5_{16}$

(b) 341_{10}

$= 155_{16}$

$$\begin{array}{r} 16 \overline{) 341} \\ \underline{16 21} - 5 \\ 1 - 5 \end{array}$$

(c) 10100010100101_2

0010	1000	1010	0101
↓	↓	↓	↓
2	8	A	5

$= 28A5_{16}$

(03)

(a)

$$\begin{array}{r} 1010101_2 \\ - 100100_2 \\ \hline 0110001_2 \\ \hline \hline \end{array}$$

(b)

$$245_8 + ABC_{16}$$

$$= 010100101_2 + 10101011100_2$$

$$= \underline{\underline{101101100001_2}}$$

(c)

$$EFA_{16} + 11010101_2$$

$$= 11101111010_2 + 11010101_2$$

$$= \underline{\underline{11111100111_2}}$$

Question 03

(01)

If we are working with a word editing software or notepad, word, power point, if we insert a number all of these numbers are treated as a character. So because of that reason that number is been stored inside the computer by using a respective ASCII value.

On the other hand whenever we are working with a software like calculator, Excel or any other software which is capable of performing arithmetic calculation those softwares are being treated numbers in a different way. So that way is, special number representation technique.

For each and every character inside of our keyboard has a respective ASCII value. Numbers also have ASCII value.

(02)

BCD (Binary Coded Decimal)

- each decimal digit is stored in four digits of binaries.
- sign of the number can't represent.
- Can't represent whether the number is positive or negative.

Packed Decimal representation

- In this representation sign can use.
- It uses the sign at the end of the binary conversion.

(03)

represent a negative number using Ones Complement Representation.

In this method when we convert decimal number to binary number have to make sure that has only 8 digits (8 bits) which represent a one byte.

Ex:-

$$\begin{aligned} 18 + 12 \\ = 18 + (-12) \end{aligned}$$

$$\begin{aligned} \rightarrow 18 &= 00010010 \\ 12 &= 00001100 \\ -12 &= 11110011 \end{aligned}$$

$$\begin{array}{r} 00010010 \\ 11110011 \\ \hline 100000101 \\ \text{---} \rightarrow +1 \\ \hline 00000110 \end{array}$$

Represent a negative number using twos Complement representation

If we get a negative number we should convert it to its binary equivalent with 8 digits. Then convert 0s into 1s and 1s into 0s. Then we should add a binary 1 to converted bit series.

Ex:-

$$\begin{aligned} 13 - 7 \\ = 13 + (-7) \end{aligned}$$

$$\begin{aligned} +13 &= 00001101 \\ +7 &= 00000111 \\ -7 &= 11110000 + 1 = 11110001 \end{aligned}$$

$$\begin{array}{r} 00001101 \\ 11110001 \\ \hline 10000010 \end{array}$$

(04)

a.

(a) $23 - 14$

$$\begin{array}{rcl} +23 & \rightarrow & 00011001 \\ +14 & \rightarrow & 00001110 \\ -14 & \rightarrow & 11110001 \\ & & \text{2's complement} \rightarrow \begin{array}{r} +1 \\ \hline 11110010 \end{array} \end{array}$$
$$\begin{array}{r} \rightarrow \quad \begin{array}{r} 00011001 \\ 11110010 \\ \hline 100001011 \end{array} \end{array}$$

b.

(b) $-56 - 12$

$$\begin{array}{rcl} 56 & \rightarrow & 00111000 \\ -56 & \rightarrow & 11000111 \\ & & \text{2's Com.} \rightarrow \begin{array}{r} +1 \\ \hline 11001000 \end{array} \end{array} \quad \left. \vphantom{\begin{array}{rcl} 56 & \rightarrow & 00111000 \\ -56 & \rightarrow & 11000111 \\ & & \text{2's Com.} \rightarrow \begin{array}{r} +1 \\ \hline 11001000 \end{array}} \right\} \begin{array}{rcl} 12 & \rightarrow & 00001100 \\ -12 & \rightarrow & 11110011 \\ & & \text{2's Com.} \rightarrow \begin{array}{r} +1 \\ \hline 11110100 \end{array} \end{array}$$

Answer : $\begin{array}{r} 11001000 \\ 11110100 \\ \hline 11011100 \end{array}$

Question 05

(01)

A network is an interconnection of set of devices (computers, servers, mainframes, network devices, peripherals, or other devices) connected to allow data sharing.

(02)

Bus topology

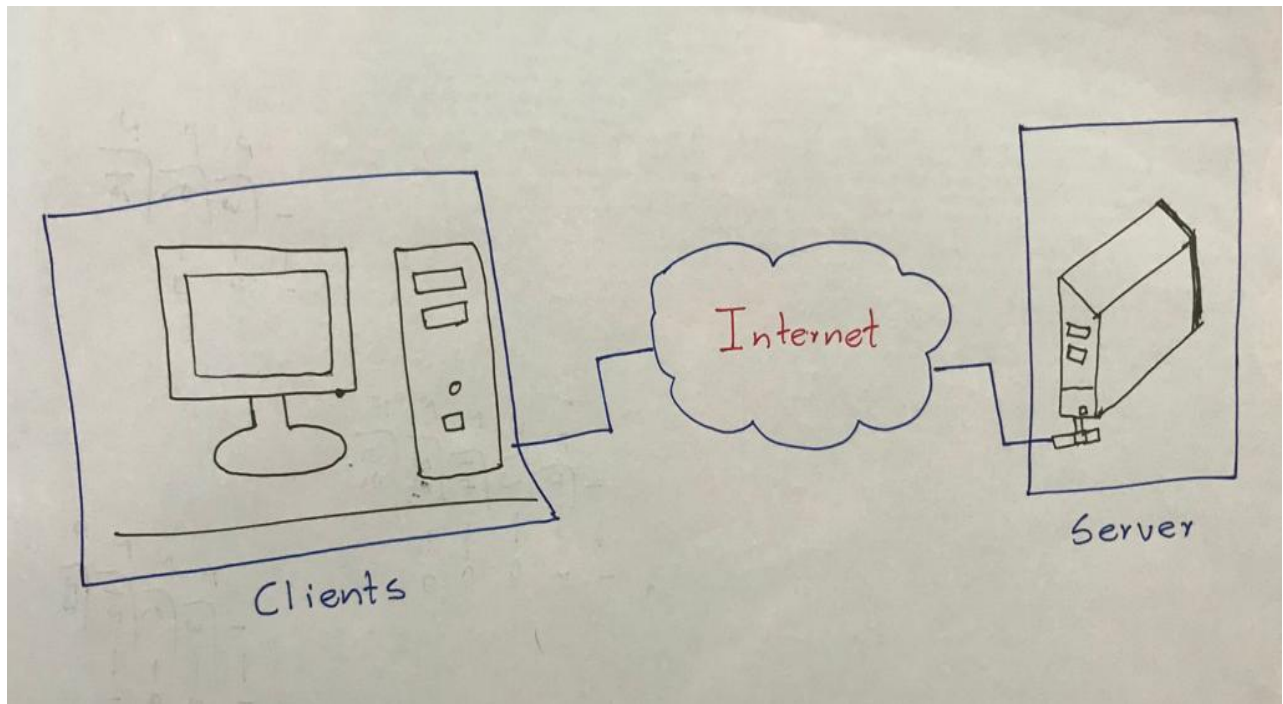
Bus topology is a network setup where each computer and network device is connected to a single cable(called a backbone).

Ring network topology

A ring topology is a network configuration where device connections create a circular data path. Each device has a dedicated point to point connection with the two devices on either side of the ring. Signals only travels in one direction it is Clockwise or Anticlockwise. When a device receives a signal intended for another device, it simply regenerates it and passes it to the next device.

(03)

In a device which used to access the website, there should be A web browsing software installed on your device which helps you to browser the web. And then we need An internet connection. It is provided by an ISP and connects you to the internet to reach to any website.



(04)

Man in the middle attacks

Man in the middle attacks are a common type of cybersecurity attack that allows attackers to eavesdrop on the communication between two targets. The attack takes place in between two legitimately communicating hosts, Once the attacker interprets the communication, they may filter and steal sensitive data and return different responses to the user.

Denial of service attack

Denial of Service attacks aims at flooding systems, networks, or servers with massive traffic, thereby making the system unable to fulfill legitimate requests. This is known as a Spread Denial of Service (DDoS) attack.

(05)

A Firewall can be simply described as a barrier built between a network of users and the external location that establishes a common security policy between the connected users and the outside world which consists of possible burglars.

Question 04

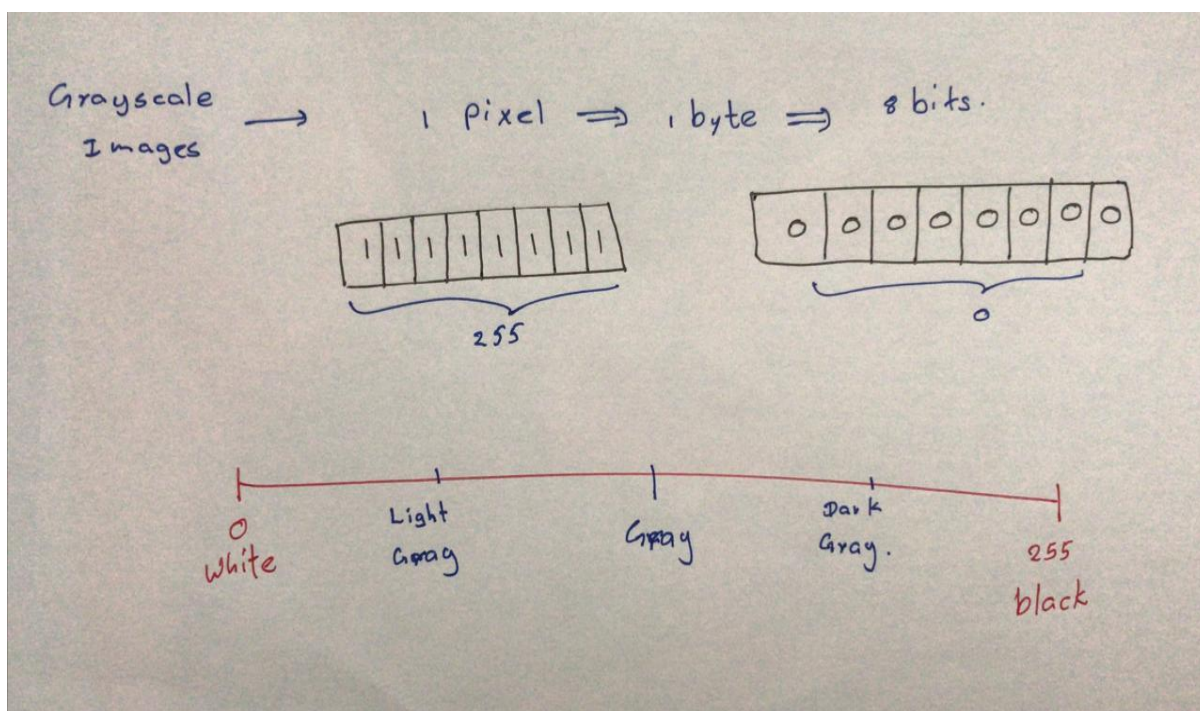
(01)

In digital imaging, a pixel picture component is the smallest addressable element in a raster image or the smallest addressable element in an all points addressable display device; so it is the smallest manageable element of a picture represented on the screen.

(02)

Gray scale image

Grayscale Images (Pixel Values = 0 – 255)



Color image

Each Pixel has three color components for example red, green, blue or (RGB) Each color component is 0 – 255.

(03)

Total resolution = 16 mega pixel

Grayscale image = 16 bytes

Colour image = 16×3 bytes

(04)

Hard disk 01

Capacity of the track = $23 \times 2\text{KB}$

Capacity of the track = $23 \times 2 \times 1000\text{KB}$

Capacity of HDD = $23 \times 2 \times 1000 \times 16\text{KB}$

Hard disk 02

Capacity of the track = $20 \times 3\text{KB}$

Capacity of the track = $20 \times 3 \times 900\text{KB}$

Capacity of HDD = $20 \times 3 \times 1000 \times 20\text{KB}$

❖ Hard disk 02 > Hard disk 01